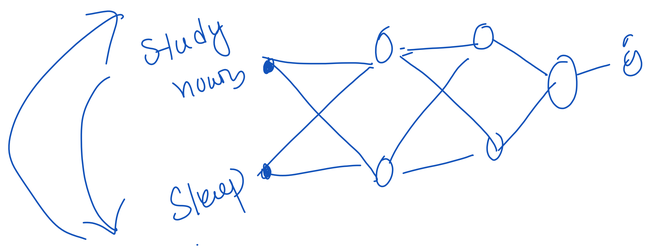
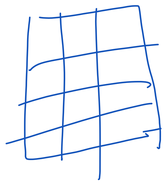
~~ANN~~ RNN

An **RNN (Recurrent Neural Network)** is a type of neural network designed to work with **sequential data**

ANN → tabular data



CNN



grid like structure

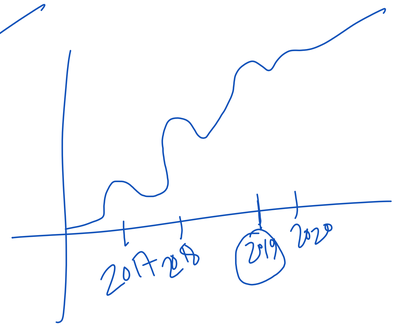
sequential data

text

Hi this Akash

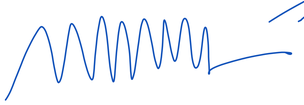
Akam Hi Thu

time series



Speech

Sequence



NLP

ANN

RNN

text: I love AI → sentiment
I hate AI

0
0
0

love, loves loving → love

vectorizer

1) number of unique words
2) 5

📝 Example Texts (3 sentences)

1. “I love AI”
2. “AI loves data”
3. “I love data”

unique words → [I love AI loves data]

[I . . . .

🔤 Step 1: Build Vocabulary (Unique Words)

List all unique words:

- I
- love
- AI
- loves
- data

👉 **Number of unique words = 5**

5

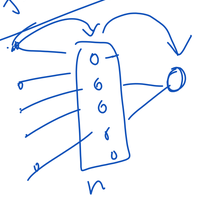
[I love AI loves data]

🔢 Step 3: One-Hot Encoding

Each word → vector of size **5** (because 5 unique words)

feed forward network

Word	One-hot vector
I	[1, 0, 0, 0, 0]
love	[0, 1, 0, 0, 0]
AI	[0, 0, 1, 0, 0]
loves	[0, 0, 0, 1, 0]
data	[0, 0, 0, 0, 1]



📊 Step 4: Encode Each Sentence

Sentence 1: “I love AI”

```
[1,0,0,0,0] - (I)
[0,1,0,0,0]   (love)
[0,0,1,0,0]   (AI)
```

Sentence 2: “AI loves data”

```
[0,0,1,0,0]   (AI)
[0,0,0,1,0]   (loves)
[0,0,0,0,1]   (data)
```

Sentence 3: “I love data”

```
[1,0,0,0,0]   (I)
[0,1,0,0,0]   (love)
```

[How RNN works]

```
[0,0,0,0,1]   (data)
```

🎯 Final Takeaways

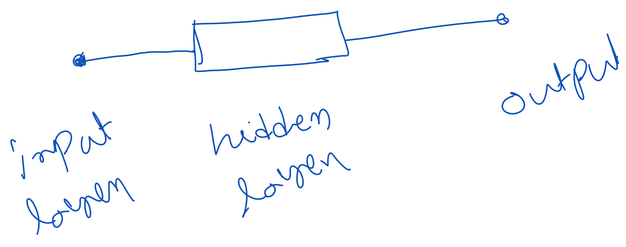
- **Unique words (vocab size) = 5**
- Each word → vector of length **5**
- One-hot encoding = **only one "1", rest "0"**

	x_{11}	x_{12}	x_{13}	Sentiment
x_1	movie	was	good	1
x_2	movie	was	bad	0
x_3	x_{31} movie	x_{32} was	x_{33} not good	0

movie was good bad not

[1 0 0 0 0], [0 1 0 0 0], [0 0 1 0 0], [0 0 0 1 0], [0 0 0 0 1]

ANN



RNN

⇒ Raw Q input (n2)
time Q, basis Q.

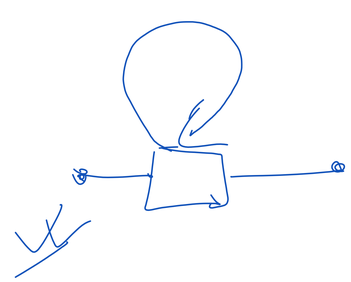
Input

x_1 x_2 x_3
movie was good → $t=1$ → movie
$t=2$ → was
$t=3$ → good

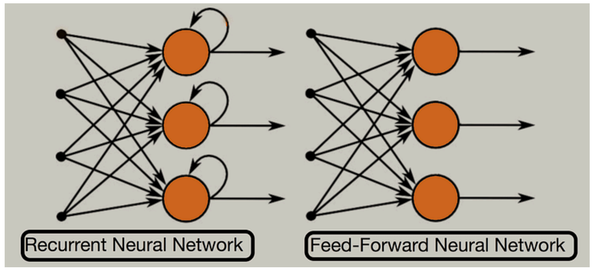
State

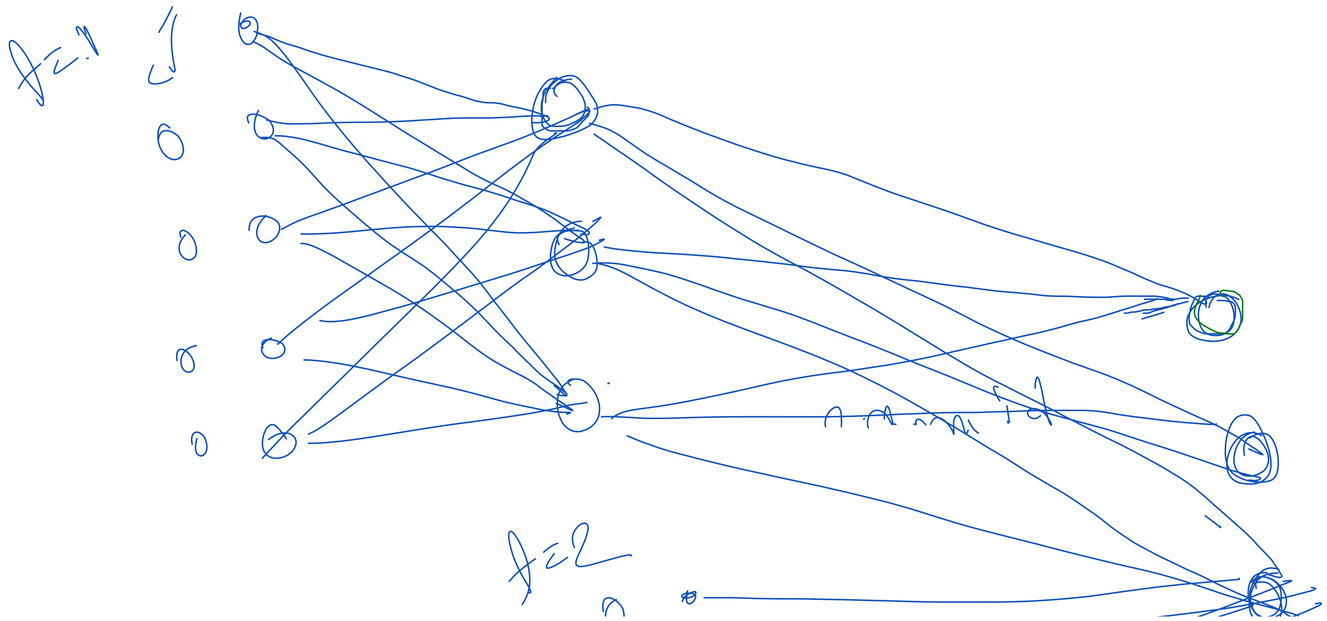
In a feedforward ANN, the entire input is processed at once, whereas an RNN processes inputs sequentially over time steps

Step	Word	Hidden State (memory)
t1	Movie	remembers "movie"
t2	was	remembers "movie was"



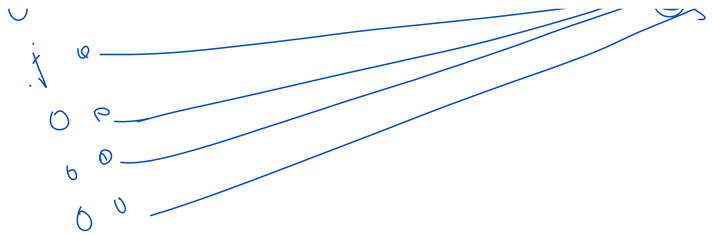
t3	not	now adds negativity
t4	good	final meaning becomes negative overall





movie was good

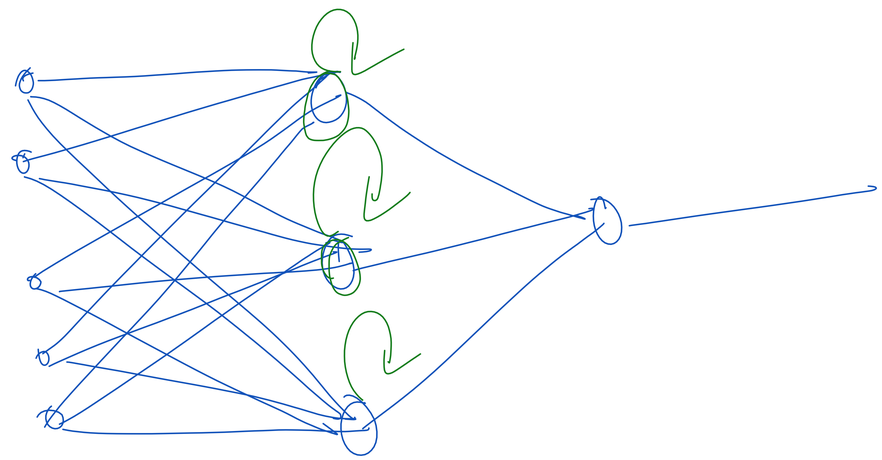
[1 0 0 0 0] [0 1 0 0 0] [0 0 1 0 0]



Current input + previous memory

feedback loop

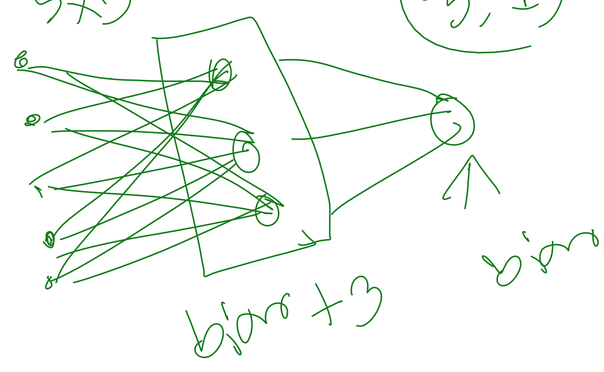
h_{t-1}



weight & bias

5x3

(3, 1)



total → 15 + 3 + 3 + 1 = 22

$2^2 + 9 = 31 + 3 =$
$= 34$

pytorch

input size = 5
hidden size = 3
batch_first = True

→ flow

class

→

- Input = movie scenes (sequence)
- RNN = watching whole movie
- Last timestep = final impression
- FC layer = decision (good or bad)
- Sigmoid = confidence (0 to 1)

```
def forward(self, x):
```

`x` = a sequence (like time steps, words, signals)

Shape: **(batch, sequence_length, features)**

1. RNN processes the sequence

```
out, _ = self.rnn(x)
```

The RNN reads the sequence **step by step**

At each timestep, it updates its memory (hidden state)

Output:

- `out` contains **information at every timestep**

2. Keep only the final understanding

```
out = out[:, -1, :]
```

👉 From all timesteps → take the **last one**

Why?

- The last hidden state = **summary of the whole sequence**

Shape becomes:

- **(batch, hidden_size)**

3. Convert understanding → prediction

```
out = self.fc(out)
```

Fully connected layer:

- Takes the learned features
- Converts them into a **single score**

4. Convert score → probability

```
out = self.sigmoid(out)
```

Sigmoid squashes value to **0–1**

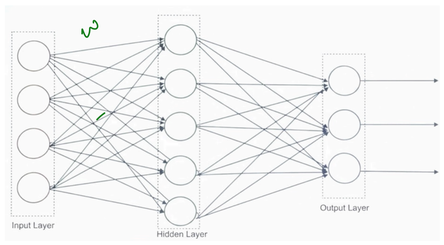
🔹 5. Return final answer

```
return out
```

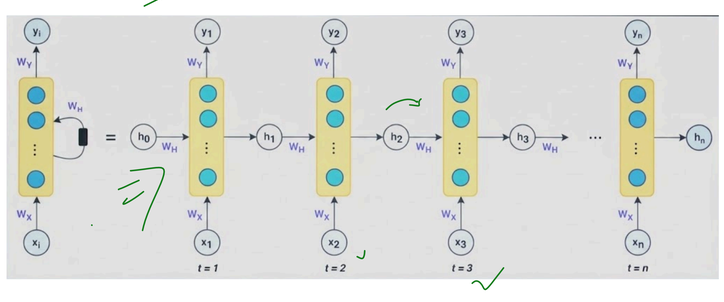
- Shape: **(batch, 1)**
- Value: probability

RNN reads the sequence → last timestep summarizes it → linear layer decides → sigmoid gives probability

ANN



Hi thro is Akram



Applications

📝 1. Text Processing (NLP)

- **Sentence prediction / next word prediction**
- Example: typing suggestions (like in Gboard)
- RNN remembers previous words to predict the next one

🌐 2. Machine Translation

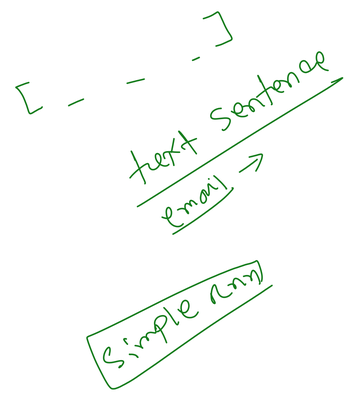
- Example: Google Translate
- Input: “I love AI” → Output: translated sentence
- RNN reads the whole sentence step-by-step and generates output

🎤 3. Speech Recognition

- Example: Siri, Google Assistant
- Converts speech → text
- RNN processes audio as a time sequence

📈 4. Time Series Prediction

- Stock prices, weather, sales forecasting



- Example: predicting tomorrow's temperature using past days
- RNN uses **previous values to predict future values**

💬 5. Chatbots

- Example: ChatGPT (earlier models used RNNs)
- Keeps track of conversation context

🎬 6. Video Processing

- Action recognition (e.g., walking, running)
- Each frame is processed sequentially

🧬 7. Healthcare

- ECG signal analysis
- Disease prediction from patient history
- RNN tracks patterns over time

✍️ 8. Text Generation

- Generating poems, stories, or code
- Example: character-level text generation